Galton–Watson branching and the constant $A(p)$

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Abstract

This chapter develops the binary Galton–Watson process and the constant $A(p)$ that governs its behaviour conditional on survival. After deriving the survival recursion and its fixed points, we turn to the subcritical range, where the population conditioned on being alive settles to a quasi-stationary law with mean $1/A(p)$ and a limiting variance determined by p and $A(p)$ alone. The constant is defined for $0 < p < \frac{1}{2}$ as the limit of $S_n/(2p)^n$; existence is established by rewriting that limit as a convergent infinite product, which also fixes the degenerate endpoint $A(0) = \frac{1}{2}$. Rescaling the survival recursion to the logistic map identifies $A(p)$ with a value of the associated Koenigs linearising function. A theorem of Becker and Bergweiler shows that this function is hypertranscendental, so that for each fixed multiplier r it has no elementary formula as a function of its spatial argument z. That theorem does not by itself settle the separate closed-form question for $p \mapsto A(p)$, but it helps explain the failure of a long search for one. We then treat the critical endpoint, where extinction is certain and the expected lifetime is nevertheless infinite, and use its power-law decay to motivate the near-critical approximation $A(p) \sim 2(1 - 2p)$. The chapter closes with a practical scheme for computing $A(p)$ across the subcritical range and a complex-plane view that places this scalar within the Koenigs coordinate; appendices preserve the full hypertranscendence proof, the complete account of the formula search, and the second-moment calculation behind the limiting conditional variance.

1 Introduction

For $0 < p < \frac{1}{2}$, both the expected population and the probability of survival decay geometrically. Their ratio is governed by a constant $A(p)$, and the population conditioned on survival consequently approaches a finite limiting mean $1/A(p)$. This chapter asks three related questions: why that constant exists, what can be said about an elementary expression for it, and how it can be computed. The answers begin with the elementary branching model and lead naturally to an infinite product, a Koenigs linearising function, and the singular limit at criticality.

The Galton–Watson process is a discrete-time, discrete-space Markov chain. It is a stochastic model in which every member of a population independently either replicates or dies at each generational increment; by these rules, the population fluctuates over the course of its lifetime. Originally formulated by Francis Galton and Henry Watson in the late nineteenth century, it stands as one of our most fundamental models of stochastic branching.

Let Z_n denote the population census at generation n. In the standard case, there is a single initial individual,

$$Z_0 = 1.$$

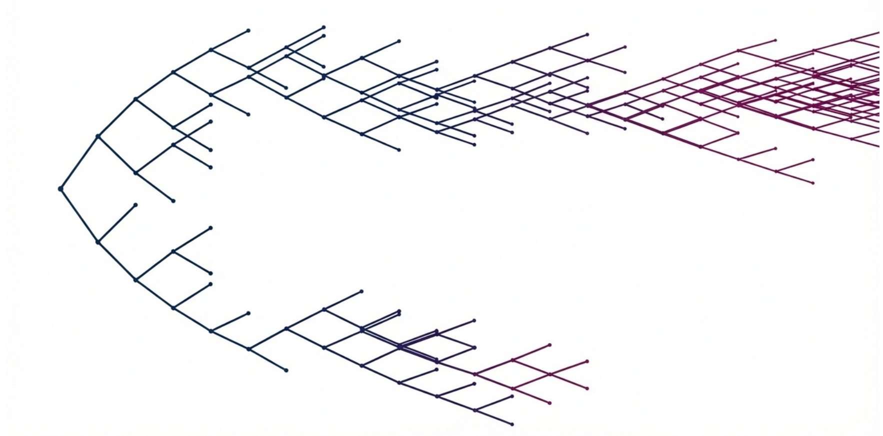
At the passage to each successive generation n, all Z_{n-1} existing individuals independently

- **replicate** with probability p, or
- **die** with the remaining likelihood $1 - p$.

This is the *simple* Galton–Watson process. It is also possible to define the process as having potentially multiple offspring per individual, but here we focus on the classic binary case.
Generally, the expected value of a discrete random variable X is defined by

$$\mathbb{E}(X) = \sum_k \mathbb{P}(X = x_k)\, x_k,$$

the sum over all possible states x_k in product with their probability of realisation.



Realisation of the simple Galton–Watson process with $p = 0.58$. Galton–Watson visualiser

Hence, after one generation (with $Z_0 = 1$), we have

$$\mathbb{E}(Z_1) = \mathbb{P}(Z_1 = 2) \cdot 2 + \mathbb{P}(Z_1 = 0) \cdot 0.$$

Replication occurs with probability p, resulting in two offspring. So this simplifies to

$$\mathbb{E}(Z_1) = 2p.$$

By the following argument, we can compute the expected population size at generation n: $\mathbb{E}(Z_n)$.

If we know the value of Z_n, then the conditional expectation of the next generation is

$$\mathbb{E}(Z_{n+1} \mid Z_n) = \sum_{i=1}^{Z_n} \mathbb{E}(Z_1) = 2p\, Z_n.$$

More generally, if we do not know Z_n, the tower property of conditional expectation gives

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}\big(\mathbb{E}(Z_{n+1} \mid Z_n)\big) = \mathbb{E}(2p\, Z_n) = 2p\, \mathbb{E}(Z_n).$$

Hence,

$$\mathbb{E}(Z_2) = 2p\, \mathbb{E}(Z_1) = (2p)^2,$$
$$\mathbb{E}(Z_3) = 2p\, \mathbb{E}(Z_2) = (2p)^3,$$

so, induction gives

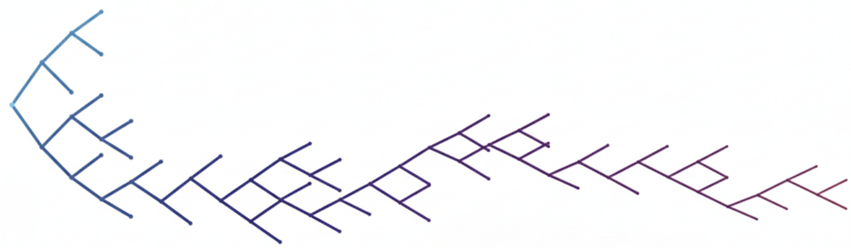
$$\mathbb{E}(Z_n) = (2p)^n.$$

From this result, we observe a standard behaviour of branching processes:

- If $2p > 1$, then $\mathbb{E}(Z_n)$ grows geometrically with n. In this case, the branching process is said to be *supercritical*.
- If $2p < 1$, then $\mathbb{E}(Z_n)$ decays geometrically with n, and the branching process is *subcritical*.

When $2p = 1$ (that is, $p = 0.5$), the Galton–Watson process is said to be *critical*. And in this case, the expectation remains constant:

$$\mathbb{E}(Z_n) = 1^n = 1 \qquad \text{for all } n.$$



Subcritical Galton–Watson realisation with $p = 0.47$.

2 Survival probability and extinction

We will see that u_n, the probability of extinction by generation n, can be computed using the probability generating function. It will follow that the probability of ultimate extinction is 1 in both critical and subcritical branching. We write $S_n = 1 - u_n$ for the survival probability, the likelihood of non-extinction at generation n. This recurrence will also provide the starting point for the constant $A(p)$ below.

The probability generating function is defined by

$$\phi(s) = \mathbb{E}(s^X),$$

for a discrete random variable X.

In our case, the offspring distribution is given by $X = Z_1$, and hence

$$\phi(s) = \mathbb{E}(s^{Z_1}) = ps^2 + (1 - p).$$

For the Galton–Watson processes, it is well known that

$$u_{n+1} = \phi(u_n),$$

where u_n denotes the probability of extinction by generation n.

This relationship can be understood intuitively by considering the fate of the population after the first generation.

- If the process dies in its first step, then extinction has already occurred. In this case, the probability of extinction by any generation is equal to 1. This event occurs with probability $1 - p$.
- If the process replicates in its first step (with probability p), then the population reaches generation 1 with two individuals. Extinction by generation $n+1$ then requires both of these fresh lineages to become extinct by their nth generation; and this happens with probability u_n^2.

Combining these cases yields the recursion

$$u_{n+1} = (1-p) + p\,u_n^2 = \phi(u_n), \tag{1}$$

which describes the probability of extinction from one generation to the next. And given the population cannot come back to life once it has died out, u_n will be a non-decreasing sequence bounded above by 1. Conversely, the probability of survival defined by the complement,

$$S_n = \mathbb{P}(Z_n > 0) = 1 - u_n,$$

will be monotonically decreasing and bounded below by 0, but may converge to a positive value $S_\infty \in [0,1]$. And to determine how these probabilities behave in the late time limit, we can use the fact that they converge and seek steady state solutions.

Substituting $u_n = 1 - S_n$ into (1), we obtain

$$1 - S_{n+1} = p(1-S_n)^2 + (1-p).$$

Expanding and simplifying gives

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1. \tag{2}$$

To find steady-state solutions, let

$$S_{n+1} = S_n = S_\infty.$$

Then S_∞ satisfies

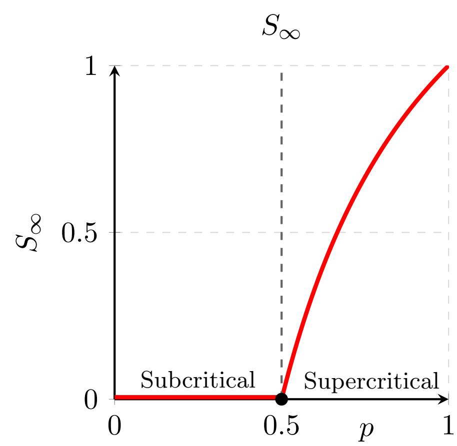
$$S_\infty = 2pS_\infty - pS_\infty^2,$$

or equivalently

$$p\,S_\infty\Big(S_\infty + \tfrac{1}{p} - 2\Big) = 0.$$

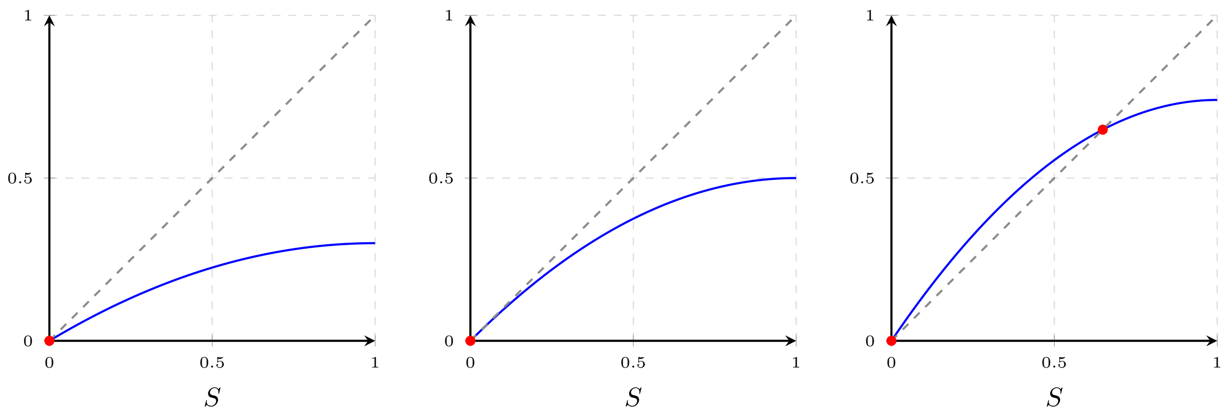
Thus there are two roots for S_∞, corresponding to two potential steady-state solutions:

$$S_\infty = 0 \quad \text{and} \quad S_\infty = 2 - \tfrac{1}{p} = \frac{2p-1}{p}.$$



But since S_n is a probability, it must lie within the interval $[0,1]$, so when $p < 0.5$, the second root is negative and thus not physically meaningful. Then, when $p > 0.5$, the probability of ultimate survival is positive, given by the larger root,

$$S_\infty = \frac{2p-1}{p}.$$



(a) Subcritical ($p = 0.3$) (b) Critical ($p = 0.5$) (c) Supercritical ($p = 0.74$)

Figure 1: Fixed points of the survival recursion $S \mapsto 2pS - pS^2$. Each panel shows the map (blue) against the diagonal (dashed), with fixed points marked in red. Below and at criticality the only fixed point in $[0, 1]$ is the origin; for $p > \frac{1}{2}$ a stable positive root $S_\infty = (2p - 1)/p$ separates from it (at $p = 0.74$, this is approximately 0.649).

So we have determined that in both the subcritical ($p < 0.5$) and critical ($p = 0.5$) cases, extinction is certain. That certainty is not the end of the matter. Below criticality a few realisations outlive the great majority, and the population conditioned on still being alive settles to a definite mean; at the critical point itself, extinction is certain and the expected lifetime is nevertheless infinite.

3 Limiting Conditional Mean $\mathbb{E}(Z_n | Z_n > 0) = 1/A(p)$

With $0 \le p \le \frac{1}{2}$, the Galton–Watson process becomes extinct almost surely. The constant $A(p)$ belongs to the subcritical range, and everything below is developed for $0 < p < \frac{1}{2}$, where $A(p) \in (0, \frac{1}{2})$. The endpoint $p = 0$ is degenerate — the process dies at the first step — and is treated separately in Section 4, where the product representation assigns it the value $A(0) = \frac{1}{2}$.

But picture this: You initiate a large number of such branching chains with the same $0 < p \le 0.5$, and watch as they simultaneously progress. Many will die out early, but some may persist for many generations. Here, we will consider the behaviour of those realisations which outlive the majority.

Specifically, we will aim to compute $\mathbb{E}(Z_n | Z_n > 0)$ for $n \to \infty$, the conditional limiting mean of non-extinct populations, and we will see that it becomes stationary.

To begin, we can decompose the expectation of population size into the two conditional expectations of persistence and cessation:

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_n | Z_n > 0) \cdot \mathbb{P}(Z_n > 0) + \mathbb{E}(Z_n | Z_n = 0) \cdot \mathbb{P}(Z_n = 0).$$

Since $\mathbb{E}(Z_n | Z_n = 0) = 0$, and $\mathbb{P}(Z_n > 0) = S_n$, this simplifies to

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_n | Z_n > 0) \cdot S_n.$$

Hence,

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{\mathbb{E}(Z_n)}{S_n}.$$

Then, we know that $\mathbb{E}(Z_n) = (2p)^n$, so

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{(2p)^n}{S_n}.$$

And we understand what happens to $(2p)^n$ for large n. But what about S_n? It may go to 0 in the limit, but how does it get there, and what will this say about the conditional mean?

Recall the recursion

$$S_{n+1} = 2p\, S_n - p\, S_n^2.$$

Because $S_n \to 0$, for growing generation count the quadratic term pS_n^2 diminishes faster than the linear term. Consequently, at large n the recursion is increasingly well approximated by

$$S_{n+1} \sim 2p\, S_n.$$

That suggests, but does not by itself establish, that S_n decays like $(2p)^n$ with some constant standing in front of it. The object we want is that constant,

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \qquad 0 < p < \tfrac{1}{2}, \tag{3}$$

and the whole content of the claim lies in the existence of the limit. The restriction to $p > 0$ is not a formality: at $p = 0$ the process dies at the first step, so $S_n = 0$ for every $n \geq 1$ and the ratio $S_n/(2p)^n$ reads $0/0$. That endpoint is defined instead by the product representation of Section 4, which gives $A(0) = \frac{1}{2}$ and agrees with the limit of $A(p)$ as $p \downarrow 0$. For $0 < p < \frac{1}{2}$, numerator and denominator both vanish, and nothing said so far prevents their ratio from drifting to 0 or growing without bound; the approximation $S_{n+1} \sim 2pS_n$ controls the increments but says nothing about what they accumulate to. So existence is not assumed here. It is established in Section 4, where the same ratio is rewritten as an infinite product whose convergence to a strictly positive value is elementary, and where the product also shows at once that $A(p) < \frac{1}{2}$. Granting it, we may write

$$S_n \sim A(p)\,(2p)^n \qquad \text{as } n \to \infty, \tag{4}$$

where the tilde means that the ratio of the two sides tends to 1, and not that the two sides are equal in the limit: S_n itself tends to 0 for $p < \frac{1}{2}$, and one must not write $\lim S_n = A(p)(2p)^n$ as an ordinary limit identity. The constant $A(p)$ absorbs the cumulative effect of the quadratic terms in the survival recurrence before the linear regime dominates — which is precisely why no argument confined to late generations will produce it. And it follows that the conditional mean in the late generation limit is given by

$$\lim_{n\to\infty} \mathbb{E}(Z_n | Z_n > 0) = \lim_{n\to\infty} \frac{(2p)^n}{S_n} = \frac{1}{A(p)}.$$

So because $A(p)$ is constant, the limiting conditional mean is also constant; it may also be referred to as the quasi-stationary mean population. The continuous-time counterpart of this constant, for the linear birth–death process, is elementary in closed form; the mathematical introduction computes it and sets the two side by side. Figure 2 shows the approach to it for five values of p, and shows at the same time the difficulty that will occupy us later: the closer p lies to $\frac{1}{2}$, the higher the plateau and the more generations the process takes to reach it. The same approach is visible in the raw paths: Figure 3 records a large ensemble at a single p close to criticality.

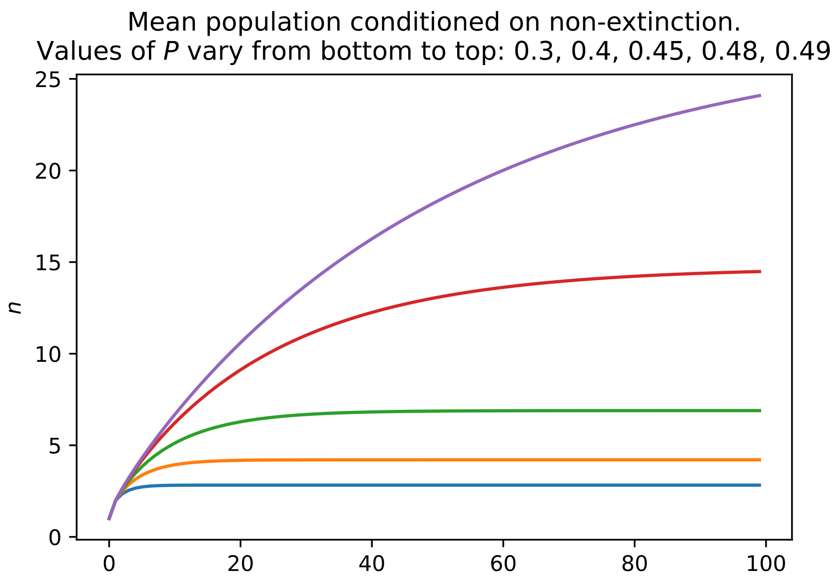


Figure 2: The conditional mean $(2p)^n/S_n$ against generation number n, with S_n obtained by iterating (2). From bottom to top, $p = 0.3, 0.4, 0.45, 0.48, 0.49$. Each curve approaches the constant $1/A(p)$; the limiting values are 2.825, 4.208, 6.893, 14.703 and 27.476. The lowest three have essentially converged by $n = 100$, but the $p = 0.49$ curve has reached only 24.1 of its limiting 27.5.

3.1 The limiting conditional variance

The mean is not the only thing that settles. Carrying the same argument through the second moment gives a limiting conditional variance

$$\mathrm{Var}(Z_n \mid Z_n > 0) \;\longrightarrow\; \frac{1}{A(p)}\left(1 + \frac{1}{1-2p}\right) - \frac{1}{A(p)^2}, \tag{5}$$

which, like the mean, is a function of p and $A(p)$ alone: conditioning on survival introduces no second unknown constant. The derivation is short and is given in Appendix C.

This is worth more than a second moment usually is. A mean that stabilises is on its own weak evidence that the conditioned population has settled down, since a mean can hold steady while the distribution behind it spreads without limit; a second moment that stabilises alongside it is the least one should ask of a population described as quasi-stationary. The law to which the conditioned population actually converges is the Yaglom limit [5], whose existence for this process is classical and is quoted from the mathematical introduction rather than re-proved here; the constant $1/A(p)$ and the expression (5) are its first two moments. Both diverge as $p \to \frac{1}{2}^-$, the variance the faster of the two, which is the quantitative form of the remark that the near-critical process is slow to settle and large when it does.

But the constant $A(p)$ is still something of a mystery. As we will see, computing an analytic formula is not straightforward.

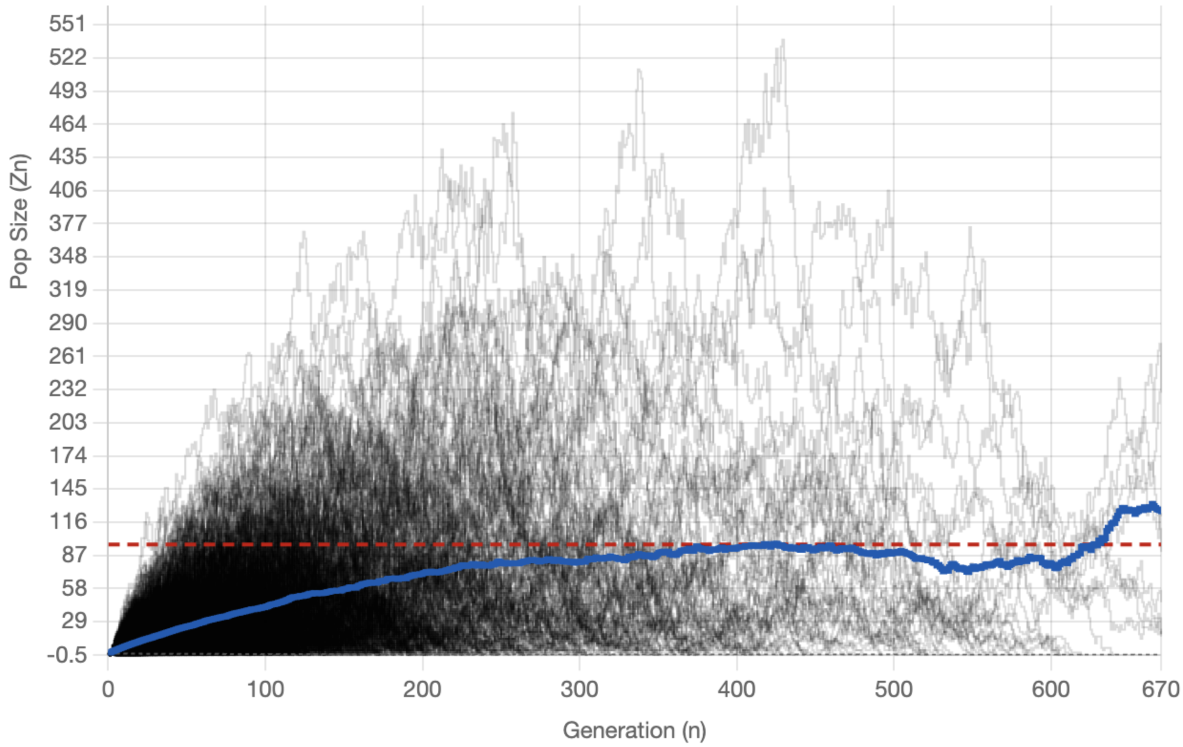


Figure 3: Ensemble of binary Galton–Watson realisations at $p = 0.49732333$, followed for 670 generations. Grey paths are individual population sizes Z_n; the blue curve is the mean among lineages still alive; the red dashed line is the limiting conditional mean $1/A(p)$. Yaglom visualiser

4 An infinite-product representation of $A(p)$

A convenient logistic-map normalisation also gives an infinite-product representation of $A(p)$. Set $S_n = 2w_n$ and $r = 2p$, so that the survival recursion (2) becomes the logistic map

$$w_{n+1} = rw_n(1 - w_n), \qquad w_0 = \tfrac{1}{2},$$

the subcritical window $0 < p < \frac{1}{2}$ corresponding to $0 < r < 1$.
Now, recall the form of $A(p)$,

$$A(p) = \lim_{n\to\infty}\left(\frac{S_n}{(2p)^n}\right) = 2\lim_{n\to\infty}\left(\frac{w_n}{r^n}\right).$$

Then, because

$$x_n = x_0\left(\frac{x_1}{x_0}\right)\left(\frac{x_2}{x_1}\right)\cdots\left(\frac{x_{n-1}}{x_{n-2}}\right)\left(\frac{x_n}{x_{n-1}}\right) = x_0\prod_{k=0}^{n-1}\frac{x_{k+1}}{x_k},$$

we can represent an infinite limit as an infinite product:

$$\lim_{n\to\infty}(x_n) = x_0\prod_{k=0}^{\infty}\left(\frac{x_{k+1}}{x_k}\right).$$

With $x_n = w_n/r^n$,

$$A(p) = 2\lim_{n\to\infty}(x_n).$$

And converting this to an infinite product as above, we can simplify the fraction,

$$\begin{aligned}\frac{x_{k+1}}{x_k} &= \frac{\frac{w_{k+1}}{r^{k+1}}}{\frac{w_k}{r^k}} = \frac{w_{k+1}}{w_k} \cdot \frac{r^k}{r^{k+1}} \\ &= \frac{rw_k(1-w_k)}{w_k} \cdot \frac{1}{r} \\ &= 1 - w_k\end{aligned}$$

leading to

$$A(p) = 2\frac{w_0}{r^0}\prod_{k=0}^{\infty}(1-w_k)$$

Then, with $w_0 = S_0/2 = 1/2$, and $r^0 = 1$, and pulling out the first ($k = 0$) factor,

$$A(p) = \frac{1}{2}\prod_{k=1}^{\infty}(1-w_k).$$

And finally, reversing the substitution of $S_n = 2w_n$,

$$A(p) = \frac{1}{2}\prod_{k=1}^{\infty}\left(1 - \frac{S_k}{2}\right). \tag{6}$$

So here we have a different formula for $A(p)$ which can be used to progressively compute an approximation by multiplying successive factors. But this still requires producing the values of S_n by iterating the nonlinear map, so it's not clear that this result brings any material advantage.

4.1 Convergence of the product

For $0 < w_k < 1$, the infinite product $\prod_{k=1}^{\infty}(1-w_k)$ converges to a strictly positive limit if and only if $\sum_{k=1}^{\infty} w_k < \infty$. From the recurrence $w_{n+1} = rw_n(1-w_n)$ with $0 < r < 1$ and $0 < w_n < 1$,

$$w_{n+1} < rw_n.$$

By induction $w_n \le w_0 r^n$. Since $0 < r < 1$, the bound is summable, so $\sum_k w_k < \infty$ and therefore

$$\prod_{k=1}^{\infty}(1-w_k)$$

converges to a positive finite value.

This settles the point left open in Section 3. The telescoping identity above is an equality at every finite n, so the limit defining $A(p)$ in (3) exists exactly when the infinite product does; the paragraph above shows that it does, and that its value is strictly positive. For $0 < p < \frac{1}{2}$ every factor $1 - S_k/2$ lies in $(0, 1)$, so (6) also gives $0 < A(p) < \frac{1}{2}$ at once, and the quasi-stationary mean $1/A(p)$ is never smaller than 2.

The product also disposes of the endpoint excluded from (3). At $p = 0$ the process dies at the first step, so $S_k = 0$ for every $k \ge 1$, every factor of (6) equals 1, and the product returns $\frac{1}{2}$. We take this as the definition,

$$A(0) = \tfrac{1}{2},$$

in place of a ratio that reads $0/0$ there. It is also the continuous extension: $S_1 = p$ and $S_{k+1} \le 2p\, S_k$ give $\sum_{k\ge1} S_k \le p/(1-2p)$, so

$$1 \;\ge\; \prod_{k=1}^{\infty}\Big(1-\frac{S_k}{2}\Big) \;\ge\; 1-\frac{p}{2(1-2p)} \;\longrightarrow\; 1 \qquad \text{as } p\downarrow 0,$$

and hence $A(p) \to \frac{1}{2}$. The endpoint value is the largest that A attains, so $1/A(0) = 2$ is the smallest quasi-stationary mean available — as it must be, since an individual that does not die leaves exactly two offspring, and therefore $Z_n \ge 2$ whenever $Z_n > 0$ and $n \ge 1$.

5 The logistic map and Koenigs representation

Recall the rescaling of Section 4: with $S_n = 2w_n$ and $r = 2p$, the survival recursion (2) is the logistic map

$$w_{n+1} = rw_n(1-w_n), \qquad w_0 = \tfrac{1}{2}, \qquad 0 < r < 1. \tag{7}$$

That the survival probabilities are the orbit of a standard dynamical system is more than a change of letters, and this section draws out what it gives: the constant $A(p)$ is the value at w_0 of the function that linearises (7).

Aside: the map outside the branching window. The logistic map is best known for its behaviour at larger multipliers, where increasing r towards 4 produces the characteristic period-doubling route to chaos (Figure 4). None of that regime is used below. The branching problem fixes $r = 2p \in (0,1)$, where the origin is attracting and the orbit from $w_0 = \frac{1}{2}$ decreases monotonically to it; the figure is included only to place the map in its usual setting.

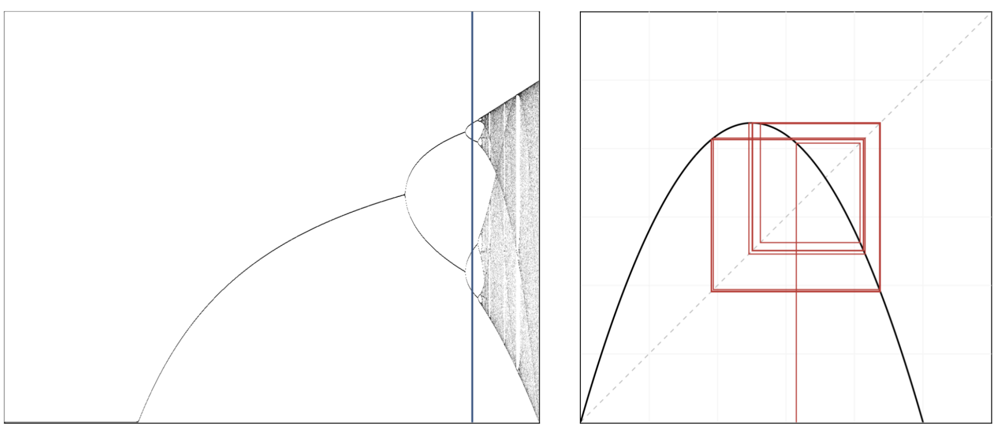


Figure 4: *Aside, not used in the argument.* Phase diagram of the logistic map that shows the period doubling of stable states, ultimately resulting in chaotic behaviour. Blue sample line on the left at $r = 3.5255$, right plot shows a trajectory of the map with the same value of r. The subcritical branching window is $r = 2p \in (0,1)$, at the far left of this diagram, where the only attractor is the fixed point at the origin. Bifurcation visualiser 👆

5.1 The identity $A(p) = 2\psi_{2p}(1/2)$

We now relate $A(p)$ to the Koenigs function ψ_r of the logistic map. The parameter r is fixed throughout this section, and the subscript is dropped where no confusion arises, so that ψ always means ψ_r.

Definition 1. *Let $f(z) = rz(1-z)$ with $0 < |r| < 1$, so that $z = 0$ is a fixed point of f with multiplier $f'(0) = r$. Koenigs proved that there is a unique function ψ_r, analytic in a neighbourhood of the origin, such that*

$$\psi_r\big(f(z)\big) = r\,\psi_r(z), \qquad \psi_r(0) = 0, \qquad \psi_r'(0) = 1.$$

The last two conditions are normalisations, fixing the origin and the scale of the new coordinate; the first says that ψ_r turns the nonlinear iteration $z \mapsto f(z)$ into multiplication by r.

Here $z = w_n$ and $f(z) = w_{n+1}$, so the functional equation reads

$$\psi(w_{n+1}) = r\,\psi(w_n),$$

which is what is meant by saying that ψ linearises the recursion. We show that $A(p) = 2\psi_r(w_0) = 2\psi_{2p}(1/2)$, and the work lies in establishing that

$$\psi(w_0) = \lim_{n\to\infty} \left(\frac{w_n}{r^n}\right).$$

Applying the functional equation at the first two steps of the orbit gives

$$\begin{aligned} &\psi(w_1) = r\psi(w_0), \\ \implies\ &\psi(w_0) = \frac{\psi(w_1)}{r}, &(8)\\ &\psi(w_2) = r\psi(w_1), \\ \implies\ &\psi(w_1) = \frac{\psi(w_2)}{r}. &(9)\end{aligned}$$

Combining (8) and (9) gives

$$\psi(w_0) = \frac{\psi(w_2)}{r^2},$$

and the same step repeated yields, by induction,

$$\psi(w_0) = \frac{\psi(w_n)}{r^n}. \tag{10}$$

Since (10) holds at every generation n, the left-hand side being independent of n, we may pass to the limit:

$$\psi(w_0) = \lim_{n\to\infty} \left(\frac{\psi(w_n)}{r^n}\right).$$

The normalisation now does the work. Definition 1 sets $\psi(0) = 0$ and $\psi'(0) = 1$, so $\psi(z)/z \to 1$ as $z \to 0$; and $w_n = S_n/2 \to 0$ as $n \to \infty$. The factor $\psi(w_n)/w_n$ therefore tends to 1, while w_n/r^n converges by Section 4, so the two limits agree:

$$\psi(w_0) = \lim_{n\to\infty} \left(\frac{w_n}{r^n}\right).$$

Since $A(p) = 2\lim_{n\to\infty}(w_n/r^n)$ and $w_0 = \frac{1}{2}$, it follows that

$$A(p) = 2\psi_r(w_0) = 2\psi_r(1/2) = 2\psi_{2p}(1/2),$$

where the subscript is a reminder that ψ_r is a different function for each value of r, and hence for each p.

5.2 The closed-form question

The identity above shows directly that $A(p)$ can be written in terms of the Koenigs function, $\psi_r(z)$, which is associated with the discrete logistic map. That does not reduce the search for a formula to a question about ψ_r — the two questions come apart, as the caveats below make clear — but it does make the analytic nature of ψ_r the natural thing to ask about next. There the answer is negative in a strong and provable sense: for every multiplier in the subcritical window the Koenigs function is not an elementary function at all.

The relevant notion is stronger than ordinary transcendence. A function g, analytic near a point, is called *differentially algebraic* if it satisfies some algebraic differential equation, meaning that there is an integer $n \geq 0$ and a non-zero polynomial P with

$$P\big(z, g(z), g'(z), \ldots, g^{(n)}(z)\big) = 0.$$

If no such equation exists, g is called *hypertranscendental*. The point of this test is its reach: every elementary function—anything assembled from algebraic, exponential, logarithmic and trigonometric pieces by finitely many arithmetic operations and compositions—is differentially algebraic. That is a classical fact rather than an observation of ours, and it rests on two ingredients. The generators of the class each satisfy an evident algebraic differential equation: an algebraic function satisfies one obtained by differentiating its defining polynomial, $g = e^z$ satisfies $g' - g = 0$, $g = \ln z$ satisfies $zg' - 1 = 0$, and $g = \sin z$ satisfies $g'' + g = 0$. And the differentially algebraic functions are closed under the arithmetic operations, differentiation and composition, so nothing built from those generators can escape the class. Rubel's survey [4] gives the closure arguments in full, together with the standard examples of functions that do escape it. So establishing hypertranscendence rules out the entire elementary class in one stroke, rather than defeating candidate formulas one at a time.

Theorem (Appendix A, Theorem 3)**.** *For every r with $0 < r < 1$, the Koenigs function ψ_r of the logistic map $f(z) = rz(1-z)$ at the fixed point $z = 0$ is hypertranscendental. Consequently it has no elementary analytic expression on any neighbourhood of the origin.*

The proof is given in Appendix A. It runs through the inverse change of coordinate: a theorem of Becker and Bergweiler [3] shows that $\theta_r = \psi_r^{-1}$, which carries the linear coordinate back to the original one, is hypertranscendental, and the fact that differential algebraicity is inherited by local inverses carries the conclusion back to ψ_r. Since the subcritical branching window $p \in (0, \frac{1}{2})$ corresponds exactly to $r = 2p \in (0, 1)$, the theorem covers every case relevant here.

Two caveats are worth stating plainly. First, the theorem concerns ψ_r as a function of z at each fixed r; it does not by itself settle whether the individual number $A(p) = 2\psi_{2p}(1/2)$, or the map $p \mapsto A(p)$, admits a closed form. Those are separate questions, and the argument here leaves them open. Second, non-elementary does not mean inaccessible: ψ_r is described exactly by its functional equation, by the limit $\lim_{n\to\infty} r^{-n} f^{\circ n}(z)$, and by a Taylor series whose coefficients follow by recursion. What is excluded is a finite elementary expression, which is precisely the thing we spent so long searching for (Appendix B).

It is not that closed forms never occur for this family. Having been discovered in the late 19th century (1884) by French mathematician Gabriel Koenigs, the function has had some time to be understood, and for the logistic map $f(z) = rz(1-z)$ it does submit to closed form description in two special cases. When $r = 2$ & $r = 4$, $\psi_r(z)$ can be written in terms of a finite number of combined elementary functions:

- $r = 2 \quad \Longrightarrow \quad \psi_2(z) = -\frac{1}{2}\ln(1-2z)$,
- $r = 4 \quad \Longrightarrow \quad \psi_4(z) = \left(\arcsin\left(\sqrt{z}\right)\right)^2$.

At $r = 2$ and $r = 4$ the multiplier at the origin is *repelling* ($|r| > 1$), so those closed forms are Poincaré-type linearisations rather than attracting-Koenigs maps for $r \in (0, 1)$. Neither case lies in the subcritical branching window $r = 2p \in (0, 1)$, and neither is a counterexample to the theorem above.

The complex-plane geometry of this coordinate is taken up in Section 9.

6 The critical endpoint: extinction and expected lifetime

We now focus on the critical case, $p = 0.5$. Although the expectation remains positive for all finite generations, ultimate extinction is certain. So the process will eventually fixate at 0, but the expected lifetime of the process is infinite.

And this is a curious mathematical result...

Intuitively, it can be understood by the fact that the total lifetime, T, of this critical branching process is distributed according to a power law.

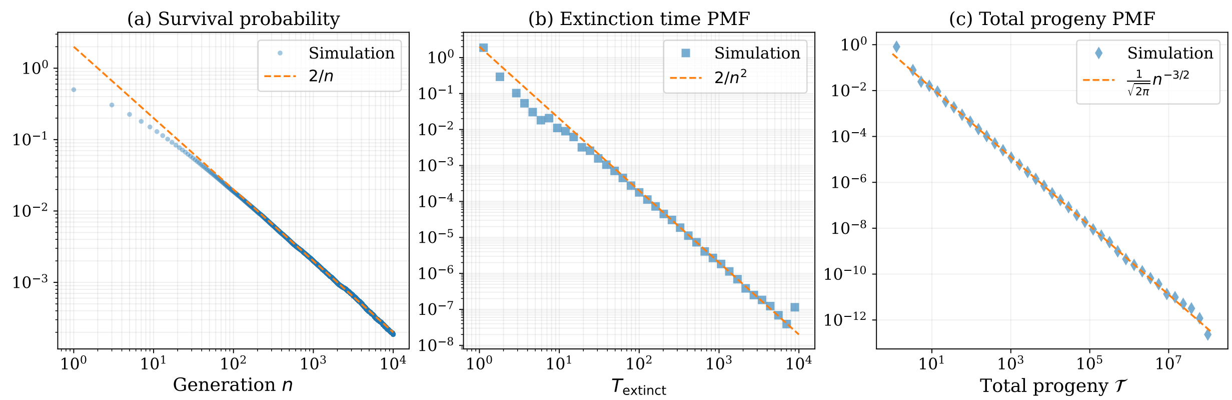


Figure 5: Power-law behaviour in a critical Galton–Watson branching process ($p = 0.5$).
(a) Survival probability $S_n = \mathbb{P}(Z_n > 0)$, showing the late generation asymptotic scaling $S_n \sim 2/n$.
(b) Probability mass function of the extinction time $T_{\text{extinct}} = \inf\{n : Z_n = 0\}$, exhibiting a heavy tail $\mathbb{P}(T_{\text{extinct}} = n) \sim 2/n^2$.
(c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} Z_k$, displaying the classical critical branching asymptotic $\mathbb{P}(\mathcal{T} = n) \sim n^{-3/2}$. ($10^6$ realisations).

For such power law distributed variables there is a modestly high chance of extraordinarily large realisations. When the expectation is infinite, the sample mean of a growing collection of realisations does not settle to any finite value. It is not that the sample average climbs steadily with sample size; it need not, and between rare events it drifts down. What happens instead is that the average is repeatedly dominated by single, exceptionally large observations: each time one arrives it lifts the average by an amount comparable to the average itself, and however many samples have already been taken, another such observation is still to come.

But aside from this intuitive understanding, we will now see how this curious result of infinite expected lifetime can be demonstrated.

6.1 Expected lifetime

We are interested in $\mathbb{E}(T)$ when $p = 0.5$, where T denotes the *extinction time* or the number of generations that the Galton Watson process survives. This is a measure of how many generations the process survives and is defined as

$$T = \inf\{n : Z_n = 0\}.$$

that is, the smallest generation index n at which the population is no longer positive. The process continues over generations

$$0, 1, 2, \ldots, T-1,$$

then fixates to 0, surviving for T generations in total.

To compute the expected time to extinction, we can entertain the following representation.

$$T = \sum_{n=0}^{\infty} \mathbf{1}_{\{T>n\}},$$

where $\mathbf{1}_{\{\text{boolean}\}}$ is the indicator function, evaluated:

$$\mathbf{1}_{\{\text{true}\}} = 1$$
$$\mathbf{1}_{\{\text{false}\}} = 0$$

The sum involving the indicator function $\mathbf{1}_{\{T>n\}}$ counts one unit for every generation in which the process is alive.

For example, if $T = 5$, then

$$\mathbf{1}_{\{T>n\}} = 1 + 1 + 1 + 1 + 1 + 0 + 0 + \cdots .$$

Then, to compute the expected value of T,

$$\mathbb{E}(T) = \mathbb{E}\left(\sum_{n=0}^{\infty} \mathbf{1}_{\{T>n\}}\right) = \sum_{n=0}^{\infty} \mathbb{E}\big(\mathbf{1}_{\{T>n\}}\big),$$

And given

$$\mathbf{1}_{\{T>n\}} = \begin{cases} 1, & \text{if } T > n, \quad (\text{alive at } n) \\ 0, & \text{otherwise,} \end{cases}$$

with

$$\mathbb{E}\big(\mathbf{1}_{\{T>n\}}\big) = \mathbb{P}(T > n).$$

$T > n$ is equivalent to $Z_n > 0$; the process is still alive at generation n. Therefore,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(Z_n > 0).$$

In other words, with $S_n = \mathbb{P}(Z_n > 0)$ being the survival probability at generation n,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n.$$

Here lies an apparent paradox. We have already established that extinction is certain in the critical case, so $S_n \to 0$ as $n \to \infty$. Yet the expected time to extinction is infinite, and $\mathbb{E}(T)$ is the sum of S_n which collapses to 0.

Explaining this will require the curious fact that the harmonic series sum is non-convergent.

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

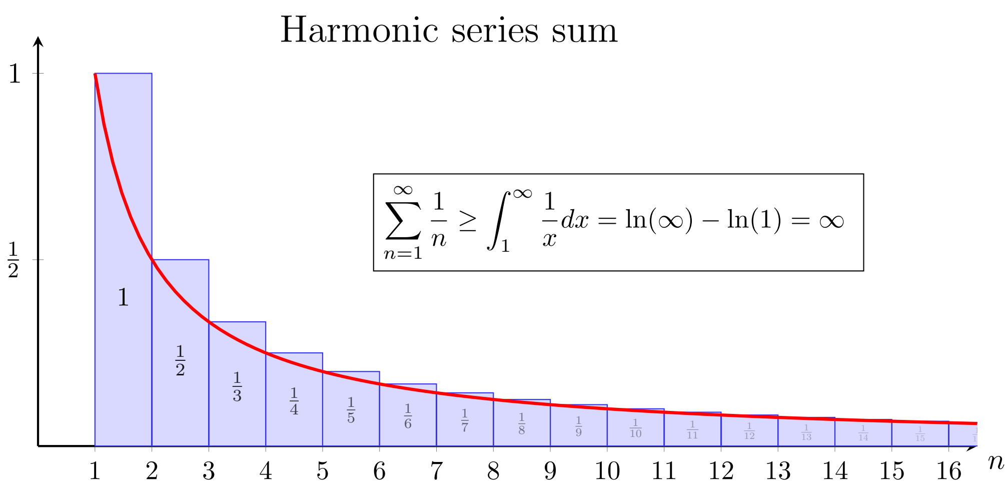


Figure 6: The harmonic series sum diverges. The sum of the areas of the blue rectangles (left) exceeds the area under the red curve (right), which diverges as $x \to \infty$. Thus, $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$. This fact will be used to show that the expected time to extinction in the critical Galton–Watson process is infinite.

To see how, we first turn our attention to the iterative formula for S_n, (2). In the critical case, $p = 0.5$, it becomes

$$S_{n+1} = S_n - \tfrac{1}{2} S_n^2 = S_n \left(1 - \frac{S_n}{2}\right). \tag{11}$$

We already know that S_n decreases to 0 here, so the reciprocal sequence $1/S_n$ increases without bound. What makes the critical case tractable is that it does so at an almost constant rate. From (11),

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{S_n \left(1 - \frac{S_n}{2}\right)} - \frac{1}{S_n} = \frac{1}{2\left(1 - \frac{S_n}{2}\right)} \longrightarrow \frac{1}{2}, \tag{12}$$

because $S_n \to 0$. A sequence whose successive increments converge to $\frac{1}{2}$ must grow like $n/2$: averaging the first n increments, which is the Stolz–Cesàro theorem and amounts to the fact that the running average of a convergent sequence tends to the same limit, gives

$$\frac{1}{n} \cdot \frac{1}{S_n} \longrightarrow \frac{1}{2},$$

and therefore

$$S_n \sim \frac{2}{n} \qquad \text{as } n \to \infty. \tag{13}$$

This is an exact asymptotic statement rather than an approximation, and it agrees with the numerical simulation results displayed in Figure 5(a).

The same result from a continuum approximation. It is worth seeing the shortcut as well, because the same device is used again in Section 7, where no such clean argument is available. Rearranging (11) gives $S_{n+1} - S_n = -\frac{1}{2}S_n^2$, and in the late generation limit, $n \gg 1$, the unit increment in n is small compared with the scale on which S_n varies. The difference on the left may then be replaced by a derivative,

$$\frac{S_{n+1} - S_n}{1} \;\to\; \frac{\mathrm{d}S(n)}{\mathrm{d}n} = -\tfrac{1}{2}S(n)^2,$$

whose general solution is

$$S(n) = \frac{2}{n + \text{const.}}$$

Nothing in that replacement justifies itself, and the constant of integration is left undetermined by it; but (13) confirms that the leading behaviour it predicts is the right one.

Then, coming back to the question of expected lifetime in the critical case, we find that the sum tends towards 2 times the harmonic series in the late generation limit:

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n}.$$

And given S_n is always positive and that the harmonic series sum diverges, it can be concluded that $E(T)$ diverges too. Thus, the expected lifetime of the critical Galton–Watson process is infinite, despite the certainty of extinction.

7 Near-critical behaviour of $A(p)$

Recall

$$S_{n+1} = 2p\, S_n - p\, S_n^2, \qquad S_0 = 1,$$

and

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{(2p)^n}{S_n} \to \frac{1}{A(p)}.$$

Direct numerical evaluation of $S_n/(2p)^n$ becomes expensive as $p \to \frac{1}{2}^-$.

Exact difference. Write $\varepsilon = 1 - 2p \to 0^+$, so $p = \frac{1}{2}(1-\varepsilon)$ and $2p = 1 - \varepsilon$. Then

$$S_{n+1} - S_n = (2p-1)S_n - p\, S_n^2 = -\varepsilon S_n - p\, S_n^2 = -\varepsilon S_n - \tfrac{1}{2}S_n^2 + \tfrac{\varepsilon}{2}S_n^2. \tag{14}$$

(The linear coefficient is $2p - 1 = -\varepsilon$.)

Continuum ODE. Treating generation number as continuous for large n and dropping $O(\varepsilon S_n^2)$ yields

$$\frac{\mathrm{d}S}{\mathrm{d}n} = -\varepsilon S - \tfrac{1}{2}S^2. \tag{15}$$

Set $u = 1/S$. Then $u' = \varepsilon u + \frac{1}{2}$, so

$$u(n) = C\, e^{\varepsilon n} - \frac{1}{2\varepsilon}, \qquad S(n) = \frac{1}{C\, e^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim A(p)\,(2p)^n \sim A\,e^{-\varepsilon n}$ for small ε gives $A = 1/C$. Matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$. Hence

$$A(p) \sim 2\varepsilon = 2(1-2p) = 2-4p \qquad \text{as } p \to \tfrac{1}{2}^{-}, \tag{16}$$

and $1/A(p) \sim 1/(2(1-2p))$. This is the practical surrogate near criticality. Substituting $C \sim 1/(2\varepsilon)$ back also gives the profile

$$S(n) \;\simeq\; \frac{2\varepsilon}{e^{\varepsilon n}-1}, \tag{17}$$

which reproduces the critical decay $2/n$ while $n \ll 1/\varepsilon$ and turns over into geometric decay beyond it. The crossover generation $n \sim 1/\varepsilon$ is the reason numerical work becomes expensive near criticality, as the next section makes quantitative.

8 Practical determination of $1/A(p)$ near criticality

8.1 Direct numerical approximation

Approximate values of $A(p)$ are obtained by iterating $S_{n+1} = f(S_n)$ and reading off the normalised ratio

$$R_N = \frac{S_N}{(2p)^N}, \tag{18}$$

which is the quantity that converges. It is worth being clear about what does not: S_N is not settling to a stationary value but decaying to zero, ultimately at the geometric rate $2p$, so that neither a threshold on S_N nor one on the difference $S_{N+1} - S_N$ is by itself a statement about the accuracy of R_N. The telescoping identity of Section 4 shows moreover that the ratio and the truncated product are the same sequence,

$$R_N = \frac{1}{2}\prod_{k=1}^{N-1}\left(1 - \frac{S_k}{2}\right),$$

so monitoring either one is the same computation.

The truncation error is then easy to control. Since $A(p) = R_N \prod_{k\geq N}(1 - S_k/2)$ and $S_{k+1} \leq 2p\,S_k$,

$$0 \;\leq\; 1 - \frac{A(p)}{R_N} \;\leq\; \frac{1}{2}\sum_{k\geq N} S_k \;\leq\; \frac{S_N}{2(1-2p)}. \tag{19}$$

Every factor omitted is less than 1, so R_N always overestimates $A(p)$, and (19) bounds the overestimate. This gives a stopping rule rather than a guess: for a target relative accuracy δ, iterate until

$$S_N \;\leq\; 2\delta\,(1-2p), \tag{20}$$

and the returned R_N is then within relative error δ of $A(p)$. The same tolerance transfers directly to the quasi-stationary mean, since $1/R_N$ and $1/A(p)$ differ by the same relative amount — though in absolute terms the error in $1/A$ is $\delta/A(p)$, which is large precisely where $A(p)$ is small.

What fails near criticality is not the rule but its cost. Writing $\varepsilon = 1 - 2p$ and using the near-critical profile (17), condition (20) is first met at

$$N \;\approx\; \frac{1}{\varepsilon}\ln\!\left(1 + \frac{1}{\delta}\right), \tag{21}$$

so the iteration count required for a fixed accuracy grows like $1/(1-2p)$. At $\delta = 10^{-6}$ the rule stops after 1.4×10^4 iterations at $p = 0.4995$, 1.4×10^5 at $p = 0.49995$ and 6.9×10^5 at $p = 0.49999$, in each case within a fraction of a per cent of (21). The scheme therefore does not break down at some particular p; it simply becomes more expensive than one is willing to pay for, and where that point falls depends on the accuracy demanded and the iterations available.

8.2 Product/asymptotic hybrid near criticality

In the absence of an exact formula for $A(p)$ we are left with the product and the near-critical asymptotic, and the practical question is where to change from one to the other. For most of the range between $p = 0$ and $p = 0.5$ the product settles quickly: at $\delta = 10^{-6}$ the stopping rule (20) is met after 60 iterations at $p = 0.4$ and 335 at $p = 0.48$. The two branches then pull in opposite directions as $p \to \frac{1}{2}^-$. The product remains valid, and its error remains bounded by (19), but its cost grows like $1/(1-2p)$; the surrogate $A \approx 2 - 4p$ costs nothing and improves as p increases, but at any fixed $p < \frac{1}{2}$ it carries an error that the product does not. Underlying both is the fact that $A(p) \to 0$ while $1/A(p) \to \infty$, so that a small absolute error in A becomes an $O(1)$ error in the quasi-stationary mean; it is relative accuracy that must be controlled here, which is what (20) does.

The crossover should therefore be stated as a budget, not as a universal number. Fix a target relative accuracy δ for the product branch and a ceiling $N_{\max}$ on the iterations spent per value of p. By (21) the product branch stays inside that ceiling exactly while

$$1 - 2p \;\geq\; \frac{\ln(1 + 1/\delta)}{N_{\max}}, \tag{22}$$

and equality defines the crossover c. The quoted value $c = 0.49999$ is the instance $\delta = 10^{-6}$, $N_{\max} \approx 7 \times 10^5$; the distance $1 - 2c$ of the crossover from criticality scales as $1/N_{\max}$, so a tenfold budget buys one further decimal place in c.

What that choice costs is set by the surrogate on the other branch. Measured against the product, the relative error of $2 - 4p$ behaves like $\varepsilon \ln(1/\varepsilon)$ with $\varepsilon = 1 - 2p$: it is 7.7×10^{-3} at $p = 0.4995$, 1.0×10^{-3} at $p = 0.49995$ and 2.3×10^{-4} at $p = 0.49999$. Since it decreases as $p \to \frac{1}{2}^-$, its worst value on $[c, \frac{1}{2})$ is the one at the crossover itself, and the accuracy of the scheme as a whole is the larger of δ and that value. With $c = 0.49999$ the hybrid returns $A(p)$ to a relative accuracy of 10^{-6} below the crossover and about 2×10^{-4} above it. Moving c towards $\frac{1}{2}$ improves the second figure and increases the cost, and that trade, rather than any particular decimal, is what fixes the cutoff.

$$A_{\mathrm{approx}}(p) = \begin{cases} \dfrac{S_N}{(2p)^N} = \dfrac{1}{2} \displaystyle\prod_{k=1}^{N-1} \left(1 - \dfrac{S_k}{2}\right), & 0 < p \leq c, \quad N = \min\{\, n : S_n \leq 2\delta(1-2p) \,\}, \\ 2 - 4p, & c < p < \frac{1}{2}, \end{cases} \tag{23}$$

with c given by (22).

9 The linearising coordinate in the complex plane

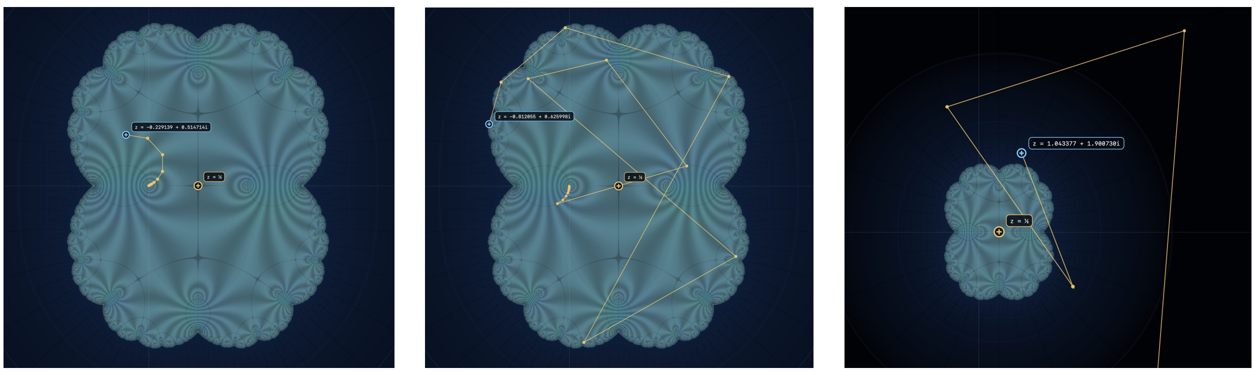
The analysis above has treated the Koenigs function ψ_r chiefly through the single value $A(p) = 2\psi_{2p}(1/2)$ and its consequences for the branching process. In the complex plane the same function can be read as a coordinate, and its geometry locates that scalar within the larger analytic object from which it comes. The figures are direct exports from the accompanying interactive program.

9.1 Basin geometry and orbit capture

For fixed r, the iteration-limit representation

$$\psi_r(z) = \lim_{n\to\infty} r^{-n} f_r^{\circ n}(z)$$

is obtained by following an orbit as it is drawn towards the origin. Its natural domain here is the attracting basin $\{z : f_r^{\circ n}(z) \to 0\}$, and this is the limit evaluated by the program. A seed may be captured within a few steps, reach the same fixed point only after a long irregular transient, or escape to infinity (Figure 7). The first two behaviours therefore belong to the same basin, however different their transients appear; the third does not. Passing from these individual orbits to the set of all captured seeds produces the coloured region in Figures 7 and 8, and in panels (a)–(e) of Figure 9. Its boundary is the Julia set of f_r [2], which separates the attracting basin from the escaping dynamics; the plotted boundary is therefore dynamical.



(a) Rapid capture. (b) Slower capture. (c) Escape to infinity.

Figure 7: Orbit fates for $f_r(z) = rz(1-z)$ at $r = 0.64$ ($p = 0.32$). The seeds $z = -0.229139 + 0.514714i$, $z = -0.812055 + 0.625998i$ and $z = 1.043377 + 1.900730i$ in (a)–(c) give, respectively, rapid capture, capture after a long transient and escape. The first two therefore lie in the same attracting basin.

9.2 The Koenigs coordinate and the constant $A(p)$

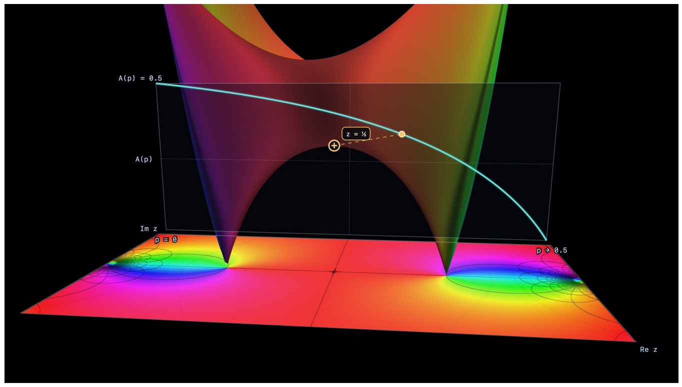
What is drawn inside the basin is the Koenigs coordinate itself. The black curves are the Cartesian grid of the linearised plane,

$$\{\operatorname{Re}\psi_r = \text{const}\}, \qquad \{\operatorname{Im}\psi_r = \text{const}\},$$

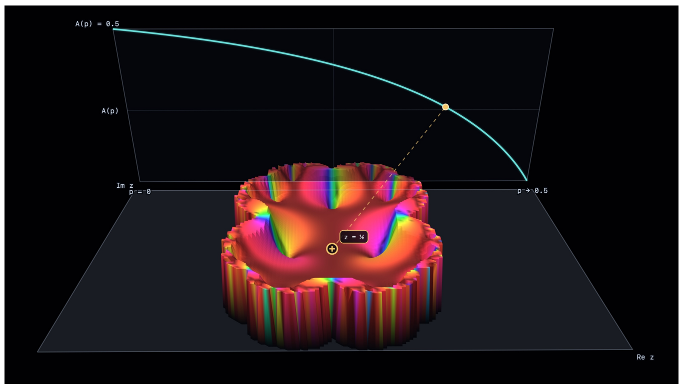
pulled back through ψ_r into the nonlinear z-plane. In the coordinate $y = \psi_r(z)$ this grid is regular and uniform; the figures record its distortion when expressed in z. The functional equation

$$\psi_r\big(f_r(z)\big) = r\psi_r(z)$$

then accounts for the nested geometry. Multiplication by r preserves the form of the linearised grid while rescaling it and, when r is complex, rotating it. Pulled back to the basin, this scaling law produces progressively smaller copies towards the attracting fixed point, with each cell recurring inside the next. The nested detail is therefore the functional equation made visible. The attracting Koenigs coordinate defined here has the basin as its natural domain; the dark exterior is instead shaded by escape rate.



(a) The height at $z = \frac{1}{2}$ against the $A(p)$ curve.



(b) The same surface viewed from lower down.

Figure 8: The Koenigs coordinate and $A(p)$ (continued on the following page).

The identity already established in Section 5,

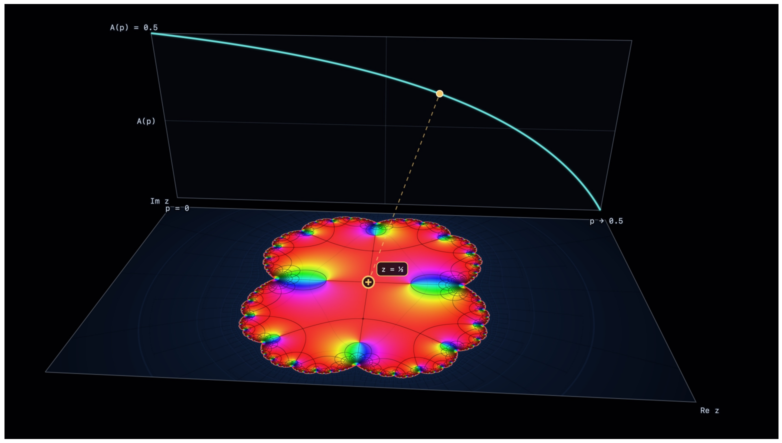
$$A(p) = 2\psi_{2p}\left(\tfrac{1}{2}\right),$$

makes Figure 8 the centre of this construction. For the real branching slice $r = 2p \in (0, 1)$, the plotted height of $|2\psi_r(z)|$ above $z = \frac{1}{2}$ is $A(p)$; its reciprocal is the quasi-stationary mean from Section 3, and its value is computed from the product of Section 4. The cyan curve on the back panel traces the same height as p ranges over $(0, \frac{1}{2})$, while the dashed line joins its value on that ordinary graph to the surface above the seed. The curve and the surface are therefore two readings of the same value.

The branching initial condition fixes the seed: $S_0 = 1$ gives $w_0 = \frac{1}{2}$. This is also the critical point of the logistic map because

$$f_r'(z) = r(1 - 2z)$$

vanishes there. The two funnels descend to the zeros $\psi_r(0) = \psi_r(1) = 0$, at the attracting fixed point and its other preimage $f_r(1) = 0$; the same two zeros flank the marker on the later coordinate plates. Thus the scalar $A(p)$ is one distinguished value of the analytic coordinate drawn across the whole basin.



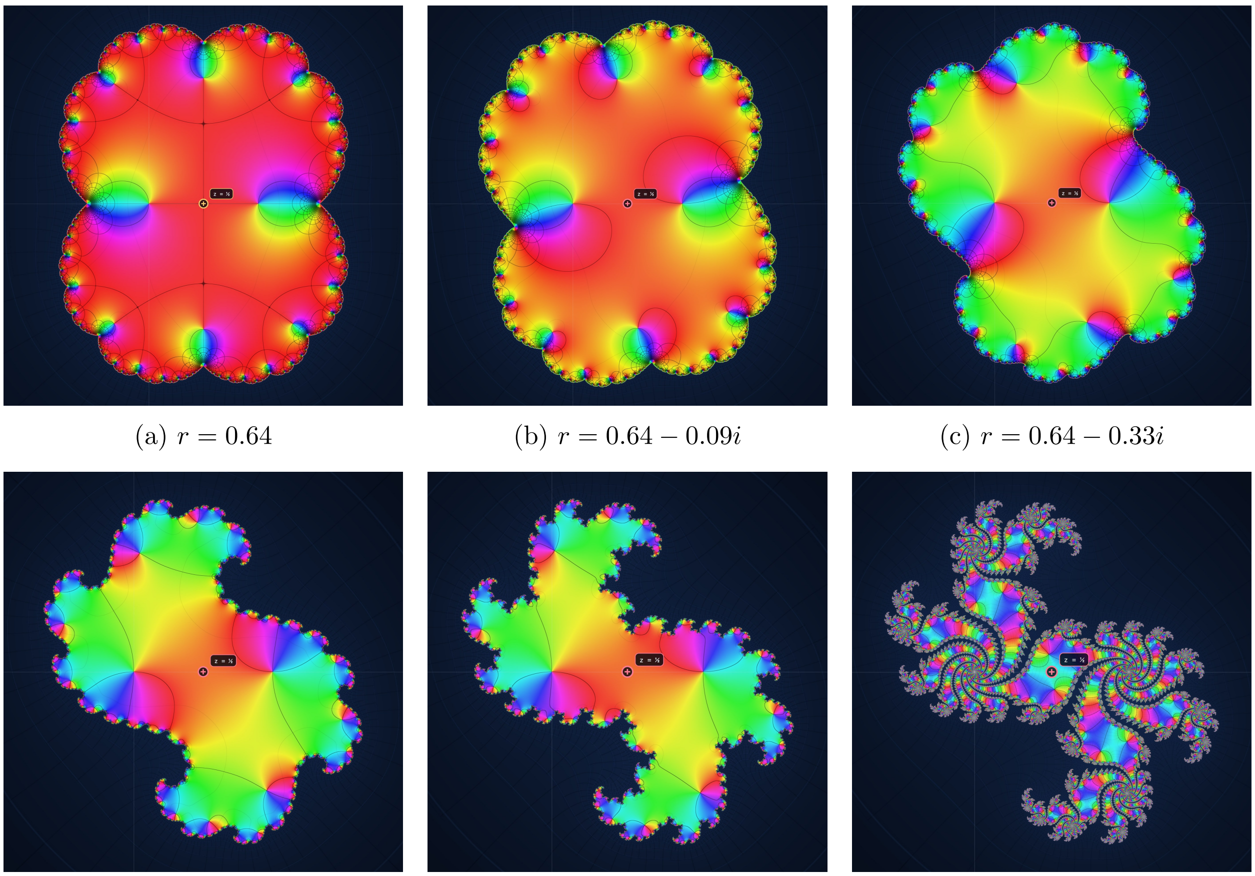
(c) The raised surface removed.

Figure 8: The Koenigs coordinate at $r = 0.64$ ($p = 0.32$). In (a), the marked seed $z = \frac{1}{2}$ has product-computed height $A(p) = 0.335521$ on $|2\psi_r(z)|$; the dashed line matches it to the cyan A-against-p curve. Panels (b) and (c) lower and remove the surface. The funnels mark $\psi_r(0) = \psi_r(1) = 0$, and the reciprocal height gives the quasi-stationary mean $1/A(p) = 2.980439$. Koenigs visualiser

9.3 Leaving the real branching slice

The branching problem occupies a one-real-dimensional slice of the two-real-dimensional r-plane. Figure 9 begins with $r = 2p = 0.64$, the branching case $p = 0.32$, and then gives r a growing imaginary part. More generally, Koenigs' construction supplies ψ_r for every multiplier with $0 < |r| < 1$, whereas the probabilistic reading requires r to lie on the open real segment $(0, 1)$. In panels (b)–(e), $|r| < 1$: the origin remains attracting, the Koenigs coordinate and its basin remain well defined, and the figures retain their complex-dynamical meaning. What is lost is the probability model. Once $r/2$ is not real in $(0, \frac{1}{2})$, it is not a branching probability, so these panels are analytic continuation in the parameter rather than new probabilistic cases. The change is in interpretation, not in the analytic construction.

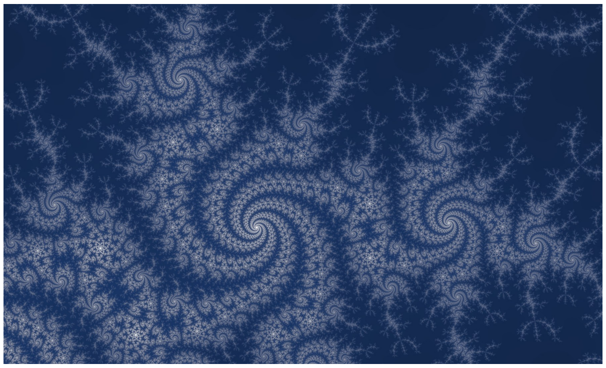
Panel (f) crosses the boundary of this construction. There $r = 0.64 - 0.78i$ has $|r| \approx 1.009 > 1$, the origin is no longer attracting, and the attracting Koenigs chart used above no longer applies.



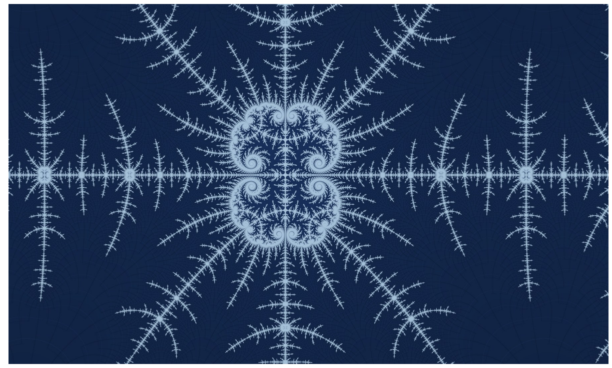
(a) $r = 0.64$ (b) $r = 0.64 - 0.09i$ (c) $r = 0.64 - 0.33i$

(d) $r = 0.64 - 0.65i$ (e) $r = 0.64 - 0.75i$ (f) $r = 0.64 - 0.78i$

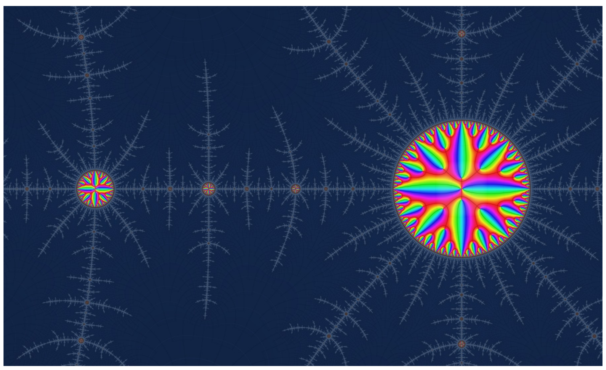
Figure 9: Continuation from the real branching slice at $\operatorname{Re} r = 0.64$. Panel (a) is the branching case $p = 0.32$, with $A(p) = 0.335521$ and $1/A(p) = 2.980439$. In (b)–(e), $|r|$ is 0.646, 0.720, 0.912 and 0.986: the Koenigs coordinate persists, but $r/2$ is no longer a probability. In (f), $|r| \approx 1.009$ and the attracting chart no longer applies. The marked point is $z = \frac{1}{2}$ throughout.



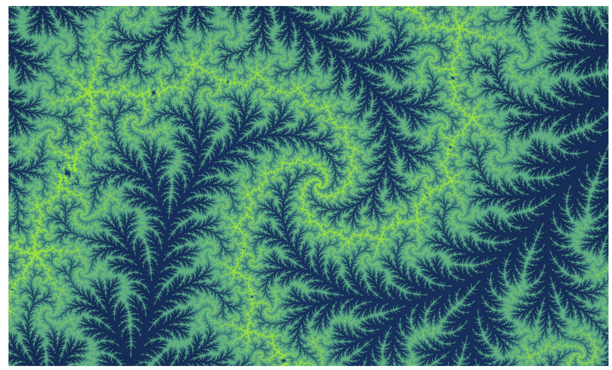
(a) $r = -0.00048 + 1.14725i$



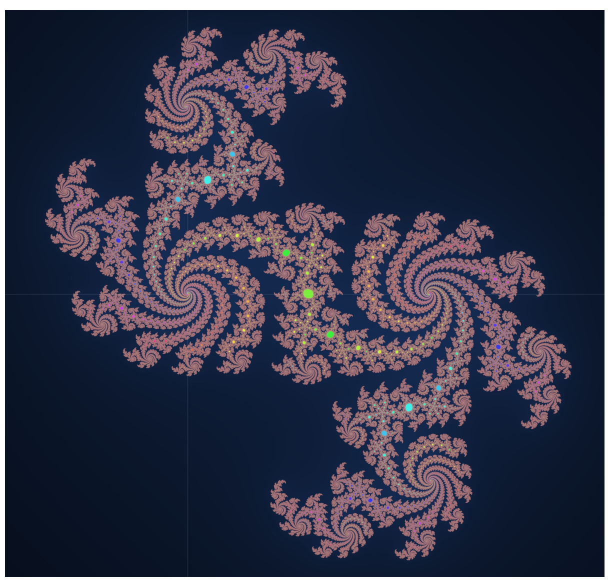
(b) $r = -1.582$



(c) $r = -1.5822$



(d) $r = -1.568 + 0.00179i$



(e) $r = 0.614 - 0.80792i$

Figure 10: Five deep zooms outside the attracting window, with $|r|$ equal to 1.147, 1.582, 1.582, 1.568 and 1.015 in panel order. Since all exceed one, none shows the attracting Koenigs chart; the images instead place the branching segment $r = 2p \in (0, 1)$ within the wider complex quadratic family. Koenigs visualiser

9.4 Beyond the attracting window

Figure 10 places the short real interval $r = 2p \in (0, 1)$ within the wider complex quadratic family. Every displayed multiplier has $|r| > 1$, so these deep zooms were generated outside the regime of the attracting Koenigs coordinate developed above. Their role is comparative: the probabilistic window is only one real segment in a parameter space that continues in both complex directions. Each image shows only a deep zoom of its coordinate plate.

The visual complexity makes the negative closed-form result of Section 5.2 feel less surprising, but it is not evidence for the theorem. The result obtained from Becker and Bergweiler concerns the function $z \mapsto \psi_r(z)$ at each fixed r and is established by the argument in Appendix A [3]; it does not decide whether the individual number $A(p)$ at any chosen parameter, or the separate map

$$p \longmapsto A(p) = 2\psi_{2p}\left(\tfrac{1}{2}\right)$$

has a closed form. Nor does a fractal basin boundary by itself preclude a tractable analytic coordinate: the Böttcher coordinate on the basin of infinity has the same Julia set as boundary and remains explicitly manageable [2]. The pictures provide context for the theorem, not a proof of it.

The complex-plane view therefore places the scalar $A(p)$ inside the geometry of the Koenigs coordinate while showing how little of the analytic family the branching model occupies. That model retains only the real segment $r = 2p \in (0, 1)$, ending at the critical boundary $r = 1$; on this slice, the distinguished coordinate value is precisely the survival constant whose probabilistic consequences were derived above.

10 Conclusion

For the subcritical binary Galton–Watson process, the survival probability has the asymptotic form

$$S_n \sim A(p)(2p)^n,$$

where $A(p)$ exists, is strictly positive, and is given by the convergent product (6). This single constant determines the first two moments of the population conditioned on survival: the mean converges to $1/A(p)$ and the variance to (5). It is therefore not merely a normalising constant for survival probability, but a quantitative descriptor of the Yaglom limit.

The logistic rescaling identifies $A(p)$ with the Koenigs-function value $2\psi_{2p}(1/2)$. Hypertranscendence rules out an elementary expression for $z \mapsto \psi_r(z)$ at fixed $r \in (0, 1)$, while leaving the separate closed-form question for $p \mapsto A(p)$ open. At the critical boundary the geometric decay is replaced by $S_n \sim 2/n$, producing infinite expected lifetime; matching the two regimes gives $A(p) \sim 2(1 - 2p)$ as $p \uparrow \frac{1}{2}$. Together, the infinite product and this near-critical asymptotic provide a practical numerical description across the subcritical interval.

A No elementary formula for the Koenigs function of the logistic map when $0 < r < 1$

For the logistic map $f_r(z) = rz(1 - z)$ with $0 < r < 1$, the origin is an attracting fixed point. Its Koenigs function ψ is a local analytic change of coordinate that turns the nonlinear iteration into the simple contraction $y \mapsto ry$. We show here that ψ satisfies no algebraic differential equation; functions with this property are called hypertranscendental. The route is indirect: a theorem of Becker and Bergweiler gives hypertranscendence for the inverse change of coordinate, and a standard fact about inverse functions carries the conclusion back to ψ. Since every elementary analytic function does satisfy an algebraic differential equation, ψ has no elementary formula near the origin. This is the result quoted in Section 5.2.

Notation. Throughout this appendix $\psi = \psi_r$ denotes the Koenigs function of Definition 1, and $\theta = \psi_r^{-1}$ its local inverse. The subscript is dropped in the content below, so ψ and θ always mean ψ_r and ψ_r^{-1}.

A.1 The question

Fix a real parameter r with $0 < r < 1$ and consider

$$f_r(z) = rz(1 - z). \tag{24}$$

In the branching setting of Section 5.2 this is the survival recurrence written in the coordinate $w_n = S_n/2$, with $r = 2p$; the subcritical window $p \in (0, \frac{1}{2})$ is exactly $r \in (0, 1)$.

The origin is a fixed point, since $f_r(0) = 0$. The derivative at a fixed point is called its *multiplier*, and here it is

$$f_r'(0) = r.$$

Because $0 < r < 1$, each application of f_r shrinks small displacements from the origin. Points that start close enough are therefore drawn towards the origin under repeated iteration, making it an *attracting* fixed point.

Koenigs' linearisation theorem [1, 2] supplies a unique function ψ, complex analytic near the origin, satisfying

$$\psi(f_r(z)) = r\psi(z), \qquad \psi(0) = 0, \qquad \psi'(0) = 1. \tag{25}$$

The last two conditions are normalisations: they place the new coordinate at zero when $z = 0$ and fix its scale. The first says that two routes lead to the same place, in that one may apply the nonlinear map f_r and then change coordinates with ψ, or change coordinates first and then simply multiply by r. In the coordinate $y = \psi(z)$, therefore, the nonlinear iteration $z \mapsto f_r(z)$ becomes the linear contraction $y \mapsto ry$.

The phrase *closed form* has no universal mathematical definition, so we pin the question down with a precise test: does ψ satisfy an algebraic differential equation? Every elementary formula satisfies some equation of this kind, so a negative answer rules out the entire class at once.

A.2 A precise test for elementary formulas

Definition 2. *An analytic function g is* differentially algebraic *if some polynomial relation holds between z, the value $g(z)$ and finitely many derivatives of g. More precisely, there must be an integer $n \geq 0$ and a non-zero polynomial P with complex coefficients such that*

$$P\big(z, g(z), g'(z), \ldots, g^{(n)}(z)\big) = 0. \tag{26}$$

If no such equation exists, g is called hypertranscendental.

Hypertranscendence is a stronger condition than ordinary transcendence. The exponential is the standard illustration: e^z is transcendental, yet $g = e^z$ satisfies $g' - g = 0$ and so is differentially algebraic. Logarithms and the trigonometric functions are differentially algebraic too, wherever they are analytic, and so is anything built from algebraic, exponential, logarithmic and trigonometric functions by finitely many arithmetic operations and compositions. To prove that ψ is hypertranscendental is therefore to prove that no expression of that kind can represent it.

The result we need is a special case of a theorem of Becker and Bergweiler [3]. Their statement concerns rational functions, so it is worth recalling that a rational function is a quotient of polynomials; every polynomial—including f_r—is therefore rational.

Theorem 1 (Becker–Bergweiler)**.** *Let R be a rational function of degree at least two with*

$$R(0) = 0, \qquad 0 < |R'(0)| < 1.$$

If a local analytic function θ satisfies

$$\theta\big(R'(0)z\big) = R(\theta(z)), \qquad \theta(0) = 0, \qquad \theta'(0) = 1, \tag{27}$$

then θ is hypertranscendental.

The role of θ is easiest to see by running the Koenigs change of coordinate backwards. Set $R = f_r$ and $\theta = \psi^{-1}$, so that $z = \theta(w)$ whenever $w = \psi(z)$. Equation (25) then reads

$$\theta(rw) = f_r(\theta(w)).$$

Because $f_r'(0) = r$, this is exactly Equation (27). Theorem 1 therefore applies to θ, the inverse Koenigs function, which carries the linear coordinate back to the original one; it does not apply to ψ directly. Terminology varies here: Becker and Bergweiler call θ a Schröder function, whereas some authors reserve that name for ψ.

One further fact lets us carry the conclusion from θ back to ψ.

Lemma 2. *Let g be analytic near the origin with $g'(0) \neq 0$. If g is differentially algebraic, then its local inverse g^{-1} is also differentially algebraic.*

Proof. Write $h = g^{-1}$, so that $g(h(w)) = w$. Differentiating this identity repeatedly, with the chain rule at each step, expresses the derivatives of g at $h(w)$ in terms of the derivatives of h. The first two are

$$g'(h(w)) = \frac{1}{h'(w)}, \qquad g''(h(w)) = -\frac{h''(w)}{(h'(w))^3}.$$

Every higher derivative has the same shape: a rational expression whose denominator is a power of $h'(w)$. Now take an equation of the form (26) for g, substitute $z = h(w)$, use $g(h(w)) = w$, and replace each derivative of g by the corresponding expression in h. Multiplying through by a sufficiently high power of $h'(w)$ clears the denominators and leaves an algebraic differential equation for h, so h is differentially algebraic. □

A.3 Main result

Theorem 3. *For every r with $0 < r < 1$, the Koenigs function ψ of $f_r(z) = rz(1-z)$ at the origin is hypertranscendental. Consequently, it has no elementary analytic expression in any neighbourhood of the origin.*

Proof. Because $\psi'(0) = 1$, the inverse-function theorem gives a local analytic inverse $\theta = \psi^{-1}$ with

$$\theta(0) = 0, \qquad \theta'(0) = 1.$$

As shown above, the Koenigs equation then reads

$$\theta(rw) = f_r(\theta(w)).$$

It remains to check the hypotheses on $R = f_r$. The map f_r is a polynomial of degree two, so it is rational and of degree at least two, and at the origin

$$f_r(0) = 0, \qquad 0 < f_r'(0) = r < 1.$$

Every hypothesis of Theorem 1 is now met, so θ is hypertranscendental.

Suppose now, for contradiction, that ψ were differentially algebraic. Lemma 2 would then make its local inverse θ differentially algebraic as well. But θ is hypertranscendental, which says precisely that it is not. This contradiction proves that ψ is hypertranscendental, and since every elementary analytic function is differentially algebraic, no elementary formula for ψ can exist locally. □

A.4 What the result means

Theorem 3 gives a rigorous sense to the informal statement that the Koenigs function has no closed form: what it rules out is an elementary formula analytic on some neighbourhood of the origin. The proof is carried out in the complex plane, but the real case is covered as well, because an elementary real-analytic formula on an interval around zero would extend locally to the same complex-analytic function.

None of this makes ψ inaccessible or uncomputable. For z sufficiently close to the origin it is given exactly by the limit

$$\psi(z) = \lim_{n\to\infty} r^{-n} f_r^{\circ n}(z), \tag{28}$$

where $f_r^{\circ n}$ denotes f_r applied n times, and its Taylor coefficients can be computed recursively from Equation (25). The functional equation, the limit and the Taylor series are all exact descriptions of ψ; what the theorem excludes is a finite elementary expression.

B Search for an elementary formula

There is no known formula which exactly describes $A(p)$, so during the project, Grant and I dedicated efforts to finding one. We tried a number of approaches; for example,

- Evaluating the $S_{n+1} = f(S_n)$ map as an ODE in the large generation limit using $(S_{n+1} - S_n)/1 \to \mathrm{d}S/\mathrm{d}n$: $\mathrm{d}S/\mathrm{d}n = (2p-1)S_n - pS_n^2$.
- A machine learning method of evolving the terms and coefficients of potential formulas using the genetic algorithm.

But unfortunately, nothing gave way to a satisfactory result. It wasn't until months after all but giving up that I decided to have another look and found the dynamical argument presented above: the Koenigs function through which $A(p)$ is expressed admits no elementary formula in its spatial argument, which is not a proof that $p \mapsto A(p)$ has none, but does say that the obvious route to one is closed.

B.1 Genetic algorithm search attempts

Before the Koenigs connection was appreciated, a number of methods were tried in search of an elementary formula. And one of those was a Genetic algorithm search based regression. The concept is simple: create 50 functions at random, measure how close each one of them is to correctly describing the curve, then select the best 5 and produce a new population of 50 by creating variations on the best 5. Measure the new 50, and repeat this cycle of variation and selection many times over. Figure 11 shows one such search, restricted to polynomials.

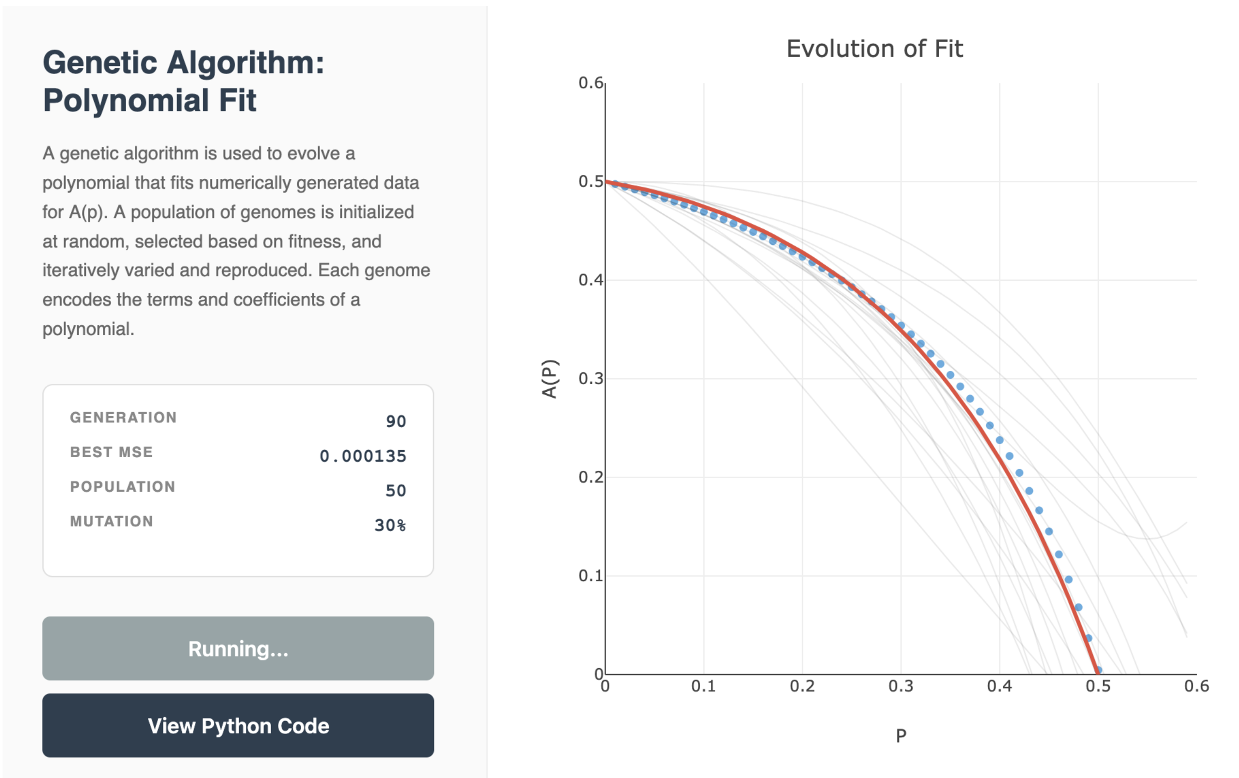


Figure 11: A genetic algorithm evolving polynomials to fit numerically generated $A(p)$ data. Each genome encodes the terms and coefficients of a polynomial. Grey curves are candidates from a population of 50 (mutation rate 30%); blue points are the data; the red curve is the best fit after 90 generations (MSE 1.35×10^{-4}). The first visualiser is a vibe-coded interface to a Python implementation I wrote by hand; v2 is a later vibe-coded attempt at a better solver, made before the Koenigs connection. Polynomial-fit visualiser 👆 Polynomial-fit visualiser (v2) 👆

When implementing the first version, I used a search space of combined polynomials; partly because I had a feeling that the solution might show up in that form, and partly because of the relative simplicity of encoding such functions.

In the minimal case, a basic polynomial of the standard form can be used.

$$a_0 + a_1 p + a_2 p^2 + a_3 p^3 + a_4 p^4 + \cdots$$

And the "genome" encoding this class of functions would be $[a_0, \ldots, a_k]$, to some arbitrarily chosen maximum order k. The initial 50 genomes are produced as lists of uniformly random generated real coefficients. It turns out that many different particular and substantially varying formulas will produce a curve that looks very close to correct, but we wanted the one exact formula... It occurred to me that the true formula may contain only integer or rational coefficients. So an updated version pursued polynomials of the form,

$$\frac{a_0}{b_0} + \frac{a_1}{b_1}p + \frac{a_2}{b_2}p^2 + \cdots + \frac{a_k}{b_k}p^k$$

with $[a_0, \ldots, a_k] \in \mathbb{Z}^k$ & $[b_0, \ldots, b_k] \in \mathbb{Z}^k$. Then, after some experiments yielding no definite solution, I also tried extending the formula space to

$$\frac{\frac{a_0}{b_0} + \frac{a_1}{b_1}p + \frac{a_2}{b_2}p^2 + \cdots + \frac{a_k}{b_k}p^k}{\frac{c_0}{d_0} + \frac{c_1}{d_1}p + \frac{c_2}{d_2}p^2 + \cdots + \frac{c_k}{d_k}p^k}$$

with genome $[a_0, \ldots, a_k, b_0, \ldots, b_k, c_0, \ldots, c_k, d_0, \ldots, d_k] \in \mathbb{Z}^{4k}$. But after numerous searches, the best that was found merely fit the high precision numerically generated data points quite well. And the formula didn't appear simple enough to represent a probably closed solution:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}$$

It started to become increasingly apparent that a huge number of very different formulas can behave almost identically in a given range.

$\hat{A}_1(p)$ does pass through (0, 0.5) & (0.5, 0) and form a pretty good approximation of the curve visually. Though it does not pass the horizontal axis with the correct gradient of -4 that we know from the asymptotic analysis above. Instead, it passes (0.5, 0) with a gradient of -3.636.

Two years after setting this work aside, I decided to have another go at using an evolutionary regression method like this one to find $A(p)$. It was just starting to become apparent how little reason there was to expect a formula, but I had another go anyway. This time with the option of using AI tools to write a more sophisticated symbolic tree-search program with leveraged time and effort. This new version is capable of searching a space of progressively nested elementary functions of all kinds. Although, of course, a similar conclusion was reached: many functions of varying form will closely resemble the target curve, but no satisfactory simple formula emerged, even on favouring solutions of less component parts.

Below are a couple of generated curves selected arbitrarily from the sea of similarly performing candidates.

$$\hat{A}_2(p) = 0.31696448624690005 \ln\Bigg(\Bigg|\cos\Bigg(\sqrt{\left|\sqrt{\left|\frac{p/p}{|3.19801133824532|}\right|}\right|} + \left(\sin\Big(\cos\big(|p|\,(0.621865696665606 - p)\,\cos(e^{e^p})\big)\Big) + e^{\ln(|p|)}\right)p\Bigg)\Bigg|\Bigg) + 0.5982282661889831$$

As you can see, there are some redundancies in the formula above such as p/p. These are introduced by the program sort of like neutral mutations in biological evolution having no effect on fitness. Below is the same formula but with these removed.

$$\hat{A}_2(p) = 0.31696448624690005\,\ln\!\left(\left|\cos\!\left(\left(\frac{1}{3.19801133824532}\right)^{1/4} + p\left[\sin\!\Big(\cos\!\big(|p|(0.621865696665606 - p)\,\cos(e^{e^p})\big)\Big) + |p|\right]\right)\right|\right) + 0.5982282661889831$$

The next one is a bit more concise:

$$\hat{A}_3(p) = \frac{-0.616647881739505\,e^{-0.552605335919693\,e^p}}{\sin(\cos p) - |p|} + 0.921964323679771$$

But a big problem with this formula and others produced by this program is that they either do not pass through the horizontal axis at (0.5,0) or they do not have the correct gradient of -4 at the critical point. And these discrepancies cause big inaccuracies when computing the quasi-stationary mean with $1/A$ as p approaches 0.5 and A draws ever closer to 0. So whilst making the program which plots many Galton-Watson trajectories along with the horizontal red dashed line indicating the QS mean, I played around with a combined approach seen in $\hat{A}_4(p)$ which aims to correct the near critical behaviour with the asymptotic formula.

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545\,e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962. \end{cases}$$

This essentially worked, but I found a better solution in a combination of the numerically generated values in combination with the asymptotic $2 - 4p$ in a narrow region of p close to 0.5.

C The limiting conditional variance

Section 3.1 quoted the limiting conditional variance (5) without deriving it. The derivation is short, and worth having: the mean of a distribution says nothing about whether the distribution itself is settling down, and a second moment that also stabilises is the least one should ask of a population claimed to be quasi-stationary.

Write ϕ_n for the probability generating function of Z_n, so that $\phi_n = \phi \circ \phi_{n-1}$, and recall that $\phi_n'(1) = \mathbb{E}(Z_n)$ and $\phi_n''(1) = \mathbb{E}\big(Z_n(Z_n - 1)\big)$. For the binary offspring law the generating function is $\phi(z) = pz^2 + (1-p)$, so its derivative is $\phi'(z) = 2pz$, and that single fact makes the chain rule collapse. Differentiating the composition once,

$$\phi_n'(z) = \phi'\big(\phi_{n-1}(z)\big)\,\phi_{n-1}'(z) = 2p\,\phi_{n-1}(z)\,\phi_{n-1}'(z), \tag{29}$$

which at $z = 1$ recovers $\mathbb{E}(Z_n) = 2p\,\mathbb{E}(Z_{n-1}) = (2p)^n$, since $\phi_{n-1}(1) = 1$. Differentiating (29) again,

$$\phi_n''(z) = 2p\Big\{\phi_{n-1}'(z)^2 + \phi_{n-1}(z)\,\phi_{n-1}''(z)\Big\}, \tag{30}$$

so, writing $v_n = \phi_n''(1)$ and evaluating at $z = 1$,

$$v_n = (2p)^{2n-1} + 2p\,v_{n-1}, \qquad v_0 = 0. \tag{31}$$

The recursion is linear and unrolls to a geometric sum,

$$v_n = \sum_{k=1}^{n} (2p)^{n-k}(2p)^{2k-1} = (2p)^n \sum_{k=1}^{n} (2p)^{k-1},$$

that is

$$\mathbb{E}\big(Z_n(Z_n-1)\big) = (2p)^n\,\frac{1-(2p)^n}{1-2p}, \qquad \text{whence} \qquad \mathbb{E}(Z_n^2) = (2p)^n\left(1+\frac{1-(2p)^n}{1-2p}\right). \tag{32}$$

The check at $n = 1$ is immediate: the right-hand side is $4p$, and Z_1 equals 2 with probability p and 0 otherwise.

Now condition. Subcriticality means $0 < 2p < 1$, so $(2p)^{2n}$ is negligible beside $(2p)^n$ at large n, and dividing (32) by $S_n \sim A(p)(2p)^n$ from (4) gives

$$\mathbb{E}(Z_n^2 \mid Z_n > 0) = \frac{\mathbb{E}(Z_n^2)}{S_n} \xrightarrow[n\to\infty]{} \frac{1}{A(p)}\left(1+\frac{1}{1-2p}\right), \tag{33}$$

the geometric factors cancelling exactly. Both conditional moments therefore converge, and subtracting the square of the first from the second gives the limiting conditional variance

$$\operatorname{Var}(Z_n \mid Z_n > 0) \;\longrightarrow\; \frac{1}{A(p)}\left(1+\frac{1}{1-2p}\right) - \frac{1}{A(p)^2}, \tag{34}$$

which is (5).

The limit (33) is again a function of p and the single constant $A(p)$ alone, so the second moment adds no new unknown.

D Near-critical growth of mean and variance

Section 3.1 noted that both limiting moments diverge as $p \to \frac{1}{2}^-$, the variance the faster of the two. The chapter's near-critical amplitude makes the rates precise. Write $\varepsilon = 1-2p$, so that $\varepsilon \downarrow 0$ as $p \uparrow \frac{1}{2}$. From

$$A(p) \sim 2\varepsilon, \qquad \frac{1}{A(p)} \sim \frac{1}{2\varepsilon}, \tag{35}$$

the limiting conditional mean is simply

$$\lim_{n\to\infty} \mathbb{E}(Z_n \mid Z_n > 0) = \frac{1}{A(p)} \sim \frac{1}{2\varepsilon}. \tag{36}$$

For the variance, rewrite (5) as

$$\frac{1}{A(p)}\left(1+\frac{1}{\varepsilon}\right) - \frac{1}{A(p)^2} = \frac{1}{A(p)\,\varepsilon} + \frac{1}{A(p)} - \frac{1}{A(p)^2}.$$

Substituting (35), the leading terms are

$$\frac{1}{A(p)\,\varepsilon} \sim \frac{1}{2\varepsilon^2}, \qquad \frac{1}{A(p)^2} \sim \frac{1}{4\varepsilon^2},$$

while $1/A(p)$ is only $O(\varepsilon^{-1})$. Hence

$$\lim_{n\to\infty} \operatorname{Var}(Z_n \mid Z_n > 0) \sim \frac{1}{4\varepsilon^2} \qquad (\varepsilon \to 0). \tag{37}$$

The mean therefore diverges like ε^{-1} and the variance like ε^{-2} — one power of ε faster, as claimed in the main text. Equivalently, if $\mu(p) = 1/A(p)$ denotes the limiting mean, then $\operatorname{Var} \sim \mu^2$, so the limiting coefficient of variation tends to 1: near criticality the quasi-stationary population is large on the scale ε^{-1}, and of relative width comparable to its mean.

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